Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with **C++** systems foundations, **simulation and integration testing** experience, and **CI/CD** discipline across cross-platform Linux and Windows environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, OOP, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
Supported design-through-deployment cycles with validation, reconciliation checks, and **CI/CD** releases across 5K+ integration jobs, maintaining 99%+ reliability in production test environments. Restructured **async orchestration** with explicit job-state transitions and failure isolation, processing 1K+ concurrent jobs and increasing throughput **3x** under load testing.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
Designed **concurrent batch-processing** with multithreading and fault-tolerant execution, handling 10K+ daily requests and 1K+ files per batch with stable production performance. Shipped validated releases with automated testing and documentation, cutting turnaround **60%** while debugging peak-load issues through logging and structured root-cause analysis.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
Built **closed-loop simulation** and experimentation pipelines over 100K+ trajectory records, enabling reproducible validation and structured test scenarios for algorithm evaluation. Produced technical reports translating simulation findings into actionable test guidance, improving cross-team alignment and engineering quality as prototypes matured toward production readiness.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
Engineered **cross-platform software** with **OOP** design patterns, automated VM builds, and integration testing across Windows, Linux, and macOS in Agile development cycles. Built 20+ unit/integration test suites with Jenkins, PyTest, and Mockito, boosting pipeline efficiency **68%** while optimizing low-level API performance **30%** through debugging and instrumentation.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
Designed cross-platform system architecture spanning 12+ backend integration points, mapping requirements into shipped changes with sub-200ms responses across 50K+ records. Troubleshot production defects via logging and automated validation, reducing manual handling **70%** while documenting interfaces and sequencing logic for cross-functional teams.

Skills

Languages: C++, Python, Java, JavaScript, TypeScript, SQL, Bash
Systems: Operating Systems, TCP/IP, Linux, Cross-Platform Development, OOP Design
Engineering: Integration Testing, Simulation, CI/CD, Jenkins, Git, Docker, Debugging
Practices: System Design, Performance Optimization, Agile, Documentation, Version Control, REST APIs, JSON, Test Automation

Projects

CPU-scheduling Simulator Live | *C++, React, OS Internals*
Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling in **C++** with edge-case handling, Gantt visualization, and turnaround-time optimization across 1K+ simulated process combinations.

Chat2SQL Code | *Java, TypeScript, Docker*
Built cross-platform integration workflows in Java and TypeScript with Docker and Maven, supporting development and deployment on **Windows, macOS, and Linux**.

Linguistix Code | *Python, AWS, Docker, CI/CD*
Deployed a containerized processing API on AWS ECS with automated **CI/CD** validation gates and production reliability checks for integration workloads.